CSCI 49376 Assignment 1

Work proposal

Big Data Technology | Two-person group

Report and code due	Sunday, October 25, 2026
Presentation	Monday, October 26, 2026
Dataset	Chosen together: cab-driver revenue, or diabetes hospital encounters
Partner A (design)	_____
Partner B (implementation)	_____

Both partners submit the same folder. On Monday, each person should be able to explain the ER diagram, the relationships, and the SQL, and to change a piece of the code if asked.

How the work is split

Partner A owns the design and the written justification. Partner B owns the database scripts, the data load, and the proof that the load worked. The dataset choice is shared, and each person reviews the other person's half before the report is merged.

Partner B can load the raw CSV into a staging table while Partner A is still writing the normalization. That step does not depend on the final table list.

Task list

Task	Owner	Hand-off
Pick the dataset and agree on the entities	Both	A short list of entities both partners accept
Profile the CSV: file name, row and column counts, what each attribute means, possible keys, missing or unknown values	A	Attribute list for the report
Normalization: original relation through 1NF and 2NF, functional dependencies, primary and foreign keys	A	Final table list B can code from
ER diagram with attributes, keys, cardinalities, and participation	A	`ER_Diagram.pdf`
Relational schema write-up, consistent with the diagram	A	Schema section of the report
`create_tables.sql`: types, keys, and `NOT NULL` or `UNIQUE` where they belong	B	A script that creates every table
`import_data.sql`: CSV into a staging table, then into each normalized table, without duplicate rows	B	A loaded database and a short description of the procedure
`verification_queries.sql` and screenshots: data is present, keys are filled in, counts look right	B	Query file and screenshots
Check that the SQL matches the diagram and the 2NF design	A	A short fix list, or confirmation that it matches
Check that the report describes the tables that were created	B	A short fix list, or confirmation that it matches
Merge the report and assemble the submission folder	Both	One shared submission
Walk through the other person's half, including one small SQL change	Both	Both partners ready for Monday

What each person should be able to explain

Partner A should be able to point at any table and say why it is in second normal form, what depends on its key, and how the relationships on the diagram become foreign keys.

Partner B should be able to run the scripts from an empty database, explain the order of the inserts, and show a query that proves a foreign key points at a real parent row.

Before the presentation, each partner reads the other half closely enough to answer a question about it and to edit a query or a `CREATE TABLE` statement on the spot.

Suggested schedule

When	What happens
Tue, Oct 20	Choose the dataset and agree on the entities
Wed, Oct 21	A finishes the profile and a first pass at 1NF and 2NF. B loads the CSV into a staging table
Thu, Oct 22	A locks the table list, keys, and ER diagram. B starts `create_tables.sql` from that list
Fri, Oct 23	B finishes the import and verification queries. A drafts the design sections of the report
Sat, Oct 24	Swap and review. Fix anything that does not match across the diagram, the schema, and the SQL
Sun, Oct 25	Merge the report, take screenshots, and submit
Mon, Oct 26	Presentation. No slides. Bring the diagram, the tables, and working SQL

Submission folder

```
CSCI49376_Assignment1_LastName/
  Assignment1_Report.pdf
  create_tables.sql
  import_data.sql
  verification_queries.sql
  ER_Diagram.pdf
```

The report covers the dataset, its size, every original attribute, the normalization steps, the functional dependencies, the final entities, all keys, the relational schema, and the import procedure. The diagram is a separate PDF. The SQL is in the three script files.